Design Specification

Autonomous Line-Following Vehicle — Track, Rules and Constraints

Revision 1.0

1. Scope

This document records the track geometry, the driving task, the lane-discipline penalties and the vehicle constraints that govern the design of the autonomous line-following vehicle. All dimensions are in millimetres unless stated otherwise.

2. Track

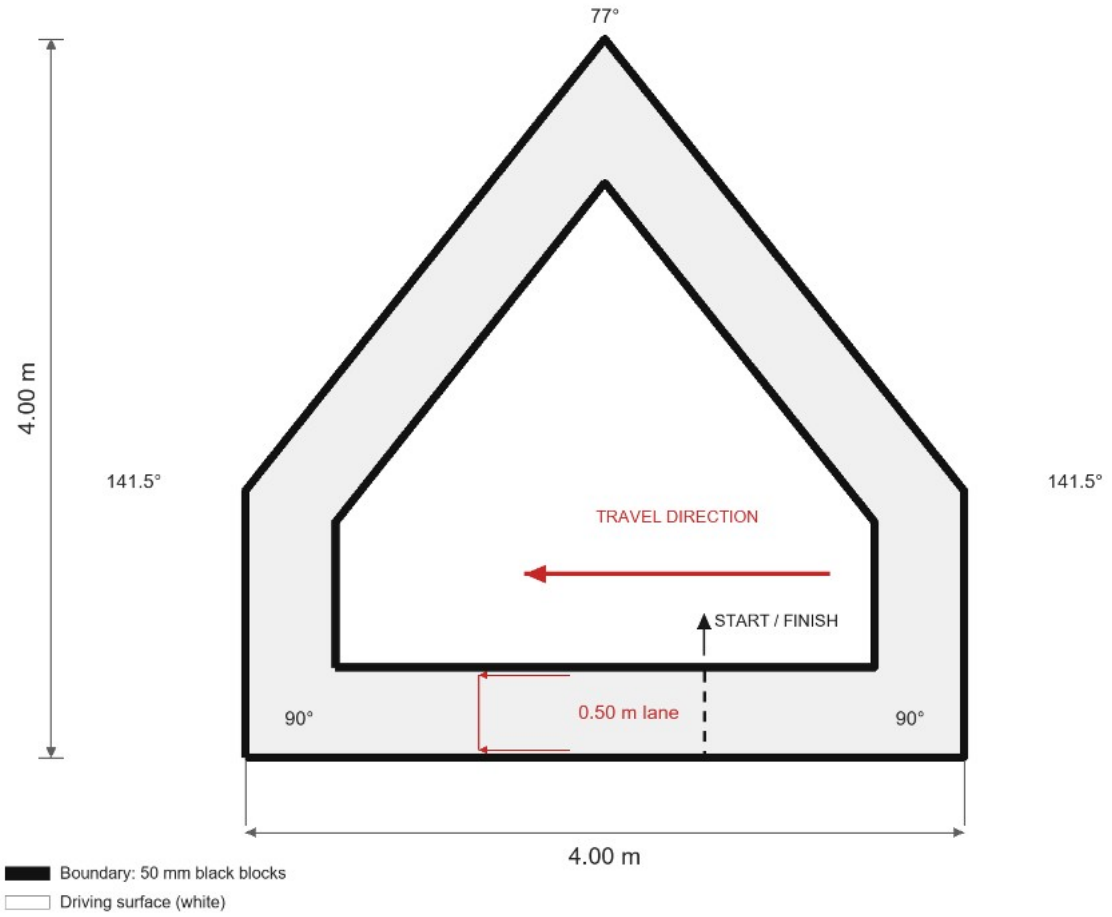


Figure 1 — Track layout (top view, not to scale)

Parameter	Value
Envelope	4.00 m × 4.00 m flat area
Lane shape	Closed loop, single lane

Parameter	Value
Lane width	500 mm (clear driving width)
Boundary marking	Continuous black blocks, 50 mm wide
Driving surface	White
Corner angles	90° ×2 (bottom), 141.5° ×2 (shoulders), 77° ×1 (apex)
Sharpest corner	77° interior angle — 103° required heading change
Start / finish	Dashed line across the lane, on the bottom straight
Direction of travel	Clockwise
Other markings	None permitted

3. Driving task

- The vehicle must complete one full clockwise lap of the closed loop automatically.
- Total lap time must not exceed 5 minutes; a shorter time scores higher.
- The vehicle is placed manually on the dashed start line, facing the direction of travel, and started with a single manual action.
- Once started, no human intervention and no remote control is permitted.
- Test runs on the track before the official measurement are allowed.

4. Lane discipline and penalties

This is the decisive rule set of the task:

Condition	Consequence
Vertical projection of the vehicle stays inside the lane at all times	No deduction
Vertical projection contacts the black boundary, once per contact	One deduction from the basic score per contact
Vertical projection is in continuous contact with the boundary for the whole lap	The basic-function score for that item is zero
Vertical projection completely leaves the lane area	The whole run is judged FAILED

In practical terms the centre line must be tracked with enough margin that neither the outer nor the inner boundary is ever touched. With a 500 mm lane and a vehicle body up to 200 mm wide, the usable lateral margin is at most 150 mm per side, and considerably less once body overhang and heading error are taken into account.

5. Vehicle constraints

Item	Limit
Length	≤ 300 mm
Width	≤ 200 mm
Height	≤ 250 mm
Total mass (including battery)	≤ 5 kg
Power source	Rechargeable battery only; no fuel, no pyrotechnics

Item	Limit
Sensors	Any type permitted

The track is laid on level ground and no marker other than the specified track markings is allowed on the driving surface. No equipment may be erected inside or around the test area, and people may walk outside the area during the run.

6. Optional voice features

Voice control is an independently scored extra function. The operator issues commands from a designated point on the field; the vehicle must respond autonomously.

Function	Requirement
Voice start	Vehicle departs on the spoken start command
Voice takeover	After crossing the finish line, a spoken command must bring the vehicle to a controlled stop within 3 seconds
Voice acknowledgement	The vehicle must reply audibly; wording is free but must be clearly intelligible
Custom commands	At least two additional commands designed by the team and demonstrated on site
Automatic stop	Stopping automatically after crossing the finish line earns bonus credit; failing to do so is not penalised

Speech recognition and speech reply must be performed on the vehicle itself. Manual buttons and wireless remote control may not be substituted for voice commands.

7. Scoring

Category	Item	Points
Report	System concept, driving and voice-interaction scheme	3
	Theoretical analysis and feasibility argument	5
	Circuit and software design	5
	Report structure and formatting	7
Basic function	One full lap, scored by lap time	60
Extra function	Voice start / takeover / reply / custom commands, automatic stop after the finish line	40
Total		120

8. Implementation notes

The accompanying firmware and vision code in this repository are built around the constraints listed above:

- Vision: 160×120 grayscale, five scan rows, per-row adaptive threshold, weighted average and a single-line fallback, so the lane centre survives the apex where the inner boundary disappears first.
- Control: cascade structure — an outer PD loop on the pixel error produces wheel speed targets, an inner PI loop on encoder counts produces PWM duty.
- Safety: if no valid frame arrives for about 120 ms the vehicle stops, so a camera failure cannot cause a run-away.
- Corner handling: the sharpest corner demands a 103° heading change with a turn radius of only 0.25 m, so the lane centre must be predicted before the apex rather than measured at it.